

SIMATS ENGINEERING

ASSESSMENT TOOL 1 - Concept Mapping

COURSE NAME: Data Structures

COURSE CODE: CSA0309

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Name	Jasmine. M

COURSE OUTCOME COVERED

CO1: Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity. (BL2, BL3)

Assessment Tool Weightage: 35%

Questions:2 EACH QUESTIONS CARRIES: 10 MARKS Total Marks:20 Duration: ONE HOUR

Q1 A smart healthcare system processes large volumes of patient data such as temperature readings, medical history, and diagnostic details. Construct a concept map linking the progression from Data → Data Object → Data Structure → Abstract Data Type (ADT) → Implementation in C. Identify relevant sub-concepts such as structures, pointers, and operations, and establish meaningful relationships among them. Organize the concept map hierarchically and justify how each level contributes to efficient data organization, processing, and improved system performance.

Q2 A smart city traffic management system handles data related to vehicle movement, road networks, and signal control. Construct a concept map connecting Primitive Data Structures → Non-Primitive Data Structures → Linear → Non-Linear Structures. Include examples such as arrays, linked lists, trees, and graphs, and illustrate their applications in traffic management. Explain the relationships between these structures and justify their classification based on functionality and efficiency.

Code of Conduct:

*I **Jasmine.M(1911260074)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



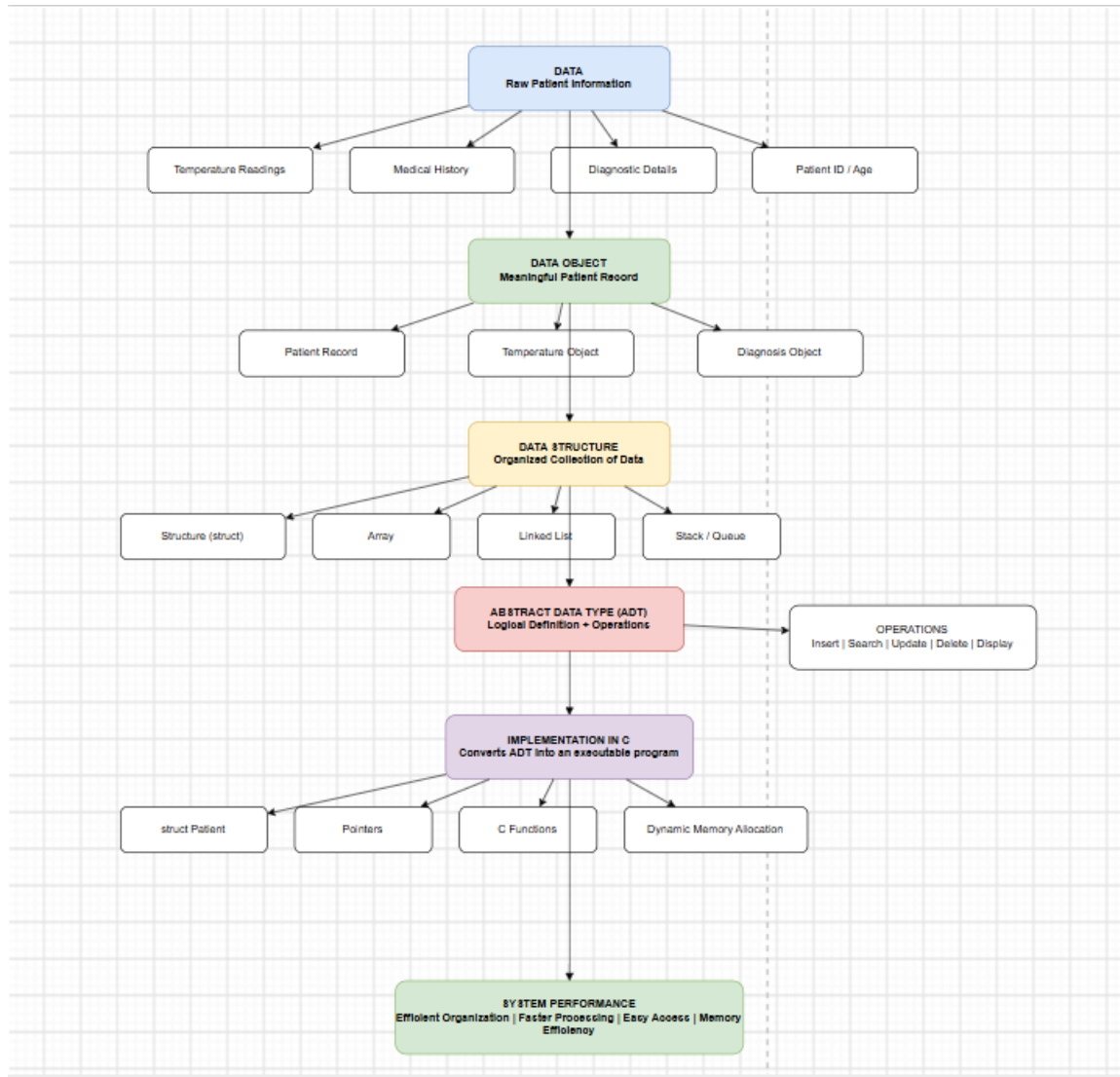
RUBRICS FOR CALCULATION:

Concept Mapping

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Q1. Smart Healthcare System: Data → Data Object → Data Structure → ADT → Implementation in C

Concept Map

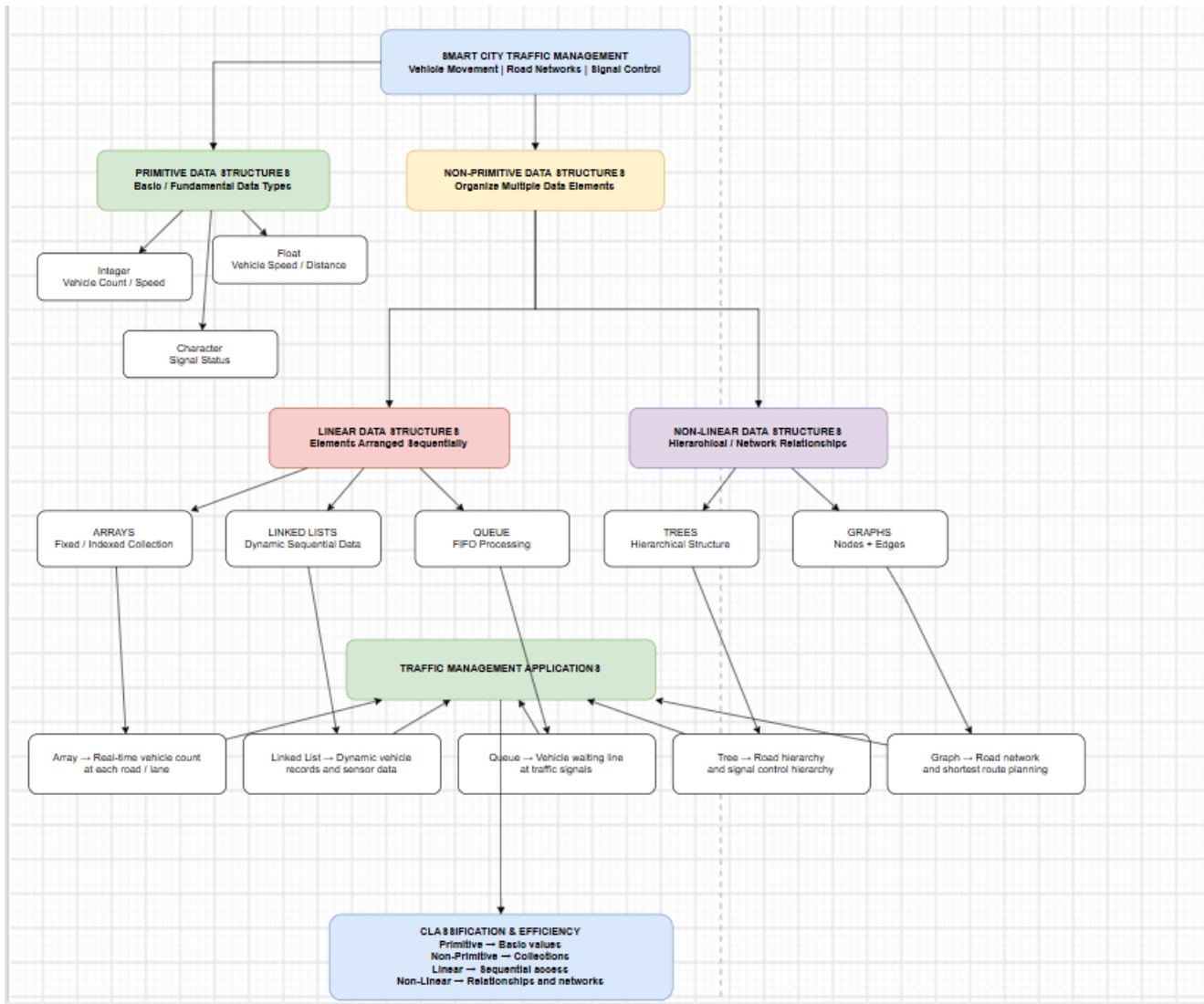


Short Description

The concept map shows how raw patient data (temperature readings, medical history, diagnostic details) is organised step by step. Data becomes a Data Object (a patient record), which is organised as a Data Structure (struct, arrays, linked lists, pointers) for storage and access. The Abstract Data Type (ADT) defines the operations (insert, delete, search, update) without exposing how they work, and the Implementation in C realises them using structures, pointers and functions. Each level adds meaning, structure, abstraction and finally working code, which gives efficient organisation, faster processing and better system performance.

Q2. Smart City Traffic Management: Primitive → Non-Primitive → Linear → Non-Linear Structures

Concept Map



Short Description

The concept map classifies the data of a smart city traffic system. Primitive data structures (int, float, char) hold simple values such as vehicle count, speed and signal status. Non-primitive structures are built from them and divided into Linear (arrays, linked lists, stacks, queues) and Non-Linear (trees, graphs). Arrays store sensor readings, linked lists and queues manage vehicle sequences and signal timing, trees organise road hierarchy, and graphs model the road network and shortest routes. They are classified by how data is arranged (sequential or hierarchical/networked) and by the efficiency of storage, access and processing.



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